D3.2 Inheritance

NAME	DATE	TOTAL
		_____ / 52

About 55 minutes, or one section at a time. A calculator is allowed. The number of marks shows how many separate points to make.

Section A: Recall

9 MARKS

1 **DEFINE** allele. [D3.2.3] [1]

2 **DISTINGUISH** between genotype and phenotype. [D3.2.3, D3.2.4] [2]

3 **STATE** the name of the enzyme that is affected in phenylketonuria. [D3.2.7] [1]

4 **STATE** the genotype or genotypes for blood groups A and O. [D3.2.9] [2]

5 **STATE** which gamete determines the sex of a zygote, and why. [D3.2.11] [2]

6 **STATE** the term for many genes affecting one characteristic. [D3.2.14] [1]

Section B: Understanding

14 MARKS

7 **EXPLAIN** why a homozygous dominant and a heterozygous organism can have the same phenotype. [D3.2.5] [2]

8 **EXPLAIN** what is meant by phenotypic plasticity, using an example. [D3.2.6] [3]

9 **EXPLAIN** how a gene pool can contain many alleles of a gene, while each individual has only two. [3]

[D3.2.8]

10 **EXPLAIN** why haemophilia is more common in males. [D3.2.12] [3]

11 **EXPLAIN** the genetic basis for rules against marriage between close relatives. [D3.2.13] [3]

Section C: Crosses and data

21 MARKS

All data in this section are invented for practice.

12 [D3.2.2, D3.2.5]

In a pea plant, tall (T) is dominant to short (t). A pure-breeding tall plant was crossed with a short plant, and the F1 plants were self-fertilized.

a **STATE** the genotype of the F1 plants. [1]

b **DRAW** a Punnett grid, and **state** the expected phenotype ratio in F2. [3]

c There were 312 tall and 96 short F2 plants. **CALCULATE** the ratio of tall to short, to one decimal place. [1]

13 [D3.2.9, D3.2.10]

A mother has blood group A and a father has blood group B. Their first child has blood group O.

a DEDUCE the genotypes of the parents. [2]

b DEDUCE the probability that their next child has group AB. [2]

c Pink Mirabilis jalapa plants are crossed together. STATE the expected ratio of flower colours. [1]

14 [D3.2.12, D3.2.13]

A woman with normal clotting has a father with haemophilia. She has children with an unaffected man.

a DEDUCE the woman's genotype, giving a reason. [2]

b DEDUCE the probability that a son of this couple is affected. [1]

15 [D3.2.14, D3.2.15]

The heights of 11 students, in cm, were: 152, 158, 160, 162, 163, 165, 166, 168, 170, 172, 190.

a STATE the median. [1]

b Q1 is 160 and Q3 is 170. CALCULATE the interquartile range, and **determine** whether 190 cm is an outlier. [3]

c DRAW a box-and-whisker plot for these data. [3]

d SUGGEST one genetic and one environmental reason for this variation. [1]

Section D: Longer answer

8 MARKS

16 **EXPLAIN** three different patterns of inheritance in which the phenotype ratio differs from a simple 3 : 1 ratio. [D3.2.10–14] [8]